IBM PROJECT REPORT

ON
predective analysis of future sales trend

Submitted By

Sadaf Abbas Zaidi

Roll No: 24051020033

Is partial fulfilment of Summer Internship for the award of the degree of

BACHELORS OF BUSINESS ADMINISTRATION (BBA - IBM)

In

Session 2026-2027

Under the Supervision of

Mr.FERHAN

(Assistant Professor)

Faculty of commerce and management

United University

Rawatpur, Prayagraj

Uttar Pradesh 211012

2. Abstract

Accurate sales forecasting is central to inventory planning, staffing and marketing budgets in retail. This project develops a predictive model for **Zenith Retail Pvt. Ltd.**, a mid-sized retail chain with four stores (Prayagraj, Lucknow, Pune and Chennai), to forecast **daily store-level sales 28 days into the future**. The dataset contains 5,859 daily records from January 2022 to December 2025, covering sales, promotions, festival periods, temperature, advertising spend and competitor events. It is a regression problem with Daily Sales (INR) as the target.

After removing 15 duplicate rows, imputing missing values and treating 14 outliers, calendar features and lag-based features (lags of 28, 35 and 56 days and a 28-day rolling mean) were engineered so that every feature is known at least 28 days in advance. Four models were trained on 2022-2024 and tested on the unseen year 2025: Linear Regression, Decision Tree, Random Forest and Gradient Boosting. All achieved an R^2 of about 0.77-0.79; **Random Forest** was selected as the best model with the lowest MAE (Rs. 4,663) and a MAPE of 6.1%. Weekday, festival periods and promotions were the strongest drivers of sales. The final model forecasts total sales of about Rs. 7.93 million for 1-28 January 2026, which supports inventory, staffing and promotion decisions.

Note: the dataset used here is simulated to resemble a real retail chain (see Section 5); the full workflow can be re-run unchanged on real company data.

3. Introduction

Domain and problem. Retail demand fluctuates with the day of week, seasons, festivals, promotions and competitor activity. Over-stocking ties up capital and causes wastage, while under-stocking means lost sales and unhappy customers. Retail chains therefore need reliable forecasts of future sales at store level.

Background. Traditional approaches (moving averages, manual estimates) cannot capture the combined effect of many drivers. Predictive analytics applies machine learning to historical data to learn these relationships and produce forecasts that can be evaluated objectively against actual results.

Importance. Better forecasts help retailers plan procurement, optimise staffing, time promotions, set sales targets and manage cash flow.

Objectives

- Build a clean, analysis-ready sales dataset and explore the main demand patterns.
- Engineer calendar and lag features suitable for a 28-day forecast horizon.
- Train and compare at least three regression models using a time-based train/test split.
- Select the best model, explain its drivers and generate a forecast for the next 28 days.
- Translate the results into actionable business recommendations.

Expected outcomes

A validated forecasting model with MAPE near 6%, a ranked list of sales drivers, a store-wise forecast for January 2026 and practical recommendations for planning.

4. Problem Statement

Zenith Retail currently plans stock and staff using last year's figures and manager judgement, which ignores promotions, festivals and recent trends. The company needs to predict **the daily sales value (INR) of each store for the next 28 days**.

Why it is required: a four-week horizon matches the supplier lead time and the monthly staffing roster. **Who benefits:** the supply-chain team (order quantities), store managers (rosters, targets), the marketing team (promotion calendar) and finance (revenue and cash-flow planning).

5. Dataset Description

Item	Details
Dataset name	zenith_retail_sales.csv (included with this submission)
Source	Simulated company dataset generated in Python (fixed random seed 42) to mimic a 4-store retail chain with trend, weekly and yearly seasonality, festivals (Holi, Diwali), promotions and realistic data-quality issues. It is **not** real company data. Real data can be substituted (e.g., Kaggle's Rossmann or Walmart sales datasets).
Period	01-Jan-2022 to 31-Dec-2025 (daily)
Size	5,859 rows x 9 columns (raw); 5,844 rows after cleaning
Target variable	Daily_Sales_INR (continuous)
Type of prediction	**Regression** (time-series forecasting framed as supervised learning)

Feature	Description
Date / Store / Region	Calendar date; one of 4 stores; North / West / South
Promotion	1 if a promotional campaign ran that day
Festival_Period	1 if within the 7 days before to 2 days after a major festival
Avg_Temp_C	Average daily temperature (deg C)
Ad_Spend_INR	Advertising spend that day
Competitor_Event	1 if a competitor ran a major sale/opening
Daily_Sales_INR	Total sales of the store that day (target)

Sample of the dataset (first 5 rows):

Date	Store	Region	Promo	Festival	Temp	Ad Spend	Comp.	Sales (INR)
2022-01-01	Chennai	South	0	0	26.0	16805.0	0	48714.92
2022-01-02	Chennai	South	0	0	22.2	13029.0	0	44815.25
2022-01-03	Chennai	South	0	0	27.1	20109.0	0	37151.31
2022-01-04	Chennai	South	0	0	23.6	13575.0	0	36967.83
2022-01-05	Chennai	South	0	0	26.1	18836.0	0	42107.16

6. Tools and Technologies Used

Category	Tool
Language	Python 3
Environment	Jupyter Notebook / Google Colab
Libraries	Pandas, NumPy (data handling); Matplotlib, Seaborn (visualisation); Scikit-learn 1.8 (models and metrics)
Dataset source	Simulated company data (Section 5)

7. Project Workflow

Step	Activity
1. Problem understanding	Define target (28-day-ahead daily sales) and stakeholders
2. Data collection	Load daily store-level records, 2022-2025
3. Data understanding and EDA	Statistics, missing values, duplicates, distributions, correlations
4. Preprocessing	Remove duplicates, impute, treat outliers, one-hot encode, split by time
5. Feature engineering	Calendar, cyclical and lag/rolling features (28-day safe)
6. Model development	Linear Regression, Decision Tree, Random Forest, Gradient Boosting
7. Model evaluation	MAE, MSE, RMSE, R^2, MAPE on the 2025 hold-out year
8. Prediction and insights	Forecast January 2026; derive business recommendations

8. Exploratory Data Analysis (EDA)

Structure and summary. The raw data has 5,859 rows and 9 columns (1 date, 2 categorical, 4 numeric/binary predictors, 1 target). Average daily sales per store are about Rs. 65,653 (std. Rs. 16,550); the maximum of Rs. 272,619 is far above the 75th percentile (Rs. 74,528), indicating outliers.

	count	mean	std	min	25%	50%	75%	max
Avg_Temp_C	5788.0	26.0	6.8	9.1	20.3	26.9	31.7	40.8
Ad_Spend_INR	5819.0	18937.7	4925.6	444.0	15680.0	18585.0	21647.5	43351.0
Daily_Sales_INR	5859.0	65653.1	16550.0	31723.0	54211.9	63472.6	74528.0	272619.2

Missing values: Avg_Temp_C has 71 missing values (1.2%) and Ad_Spend_INR has 40 (0.7%). **Duplicates:** 15 fully duplicated rows were found. Both are small and were corrected in preprocessing.

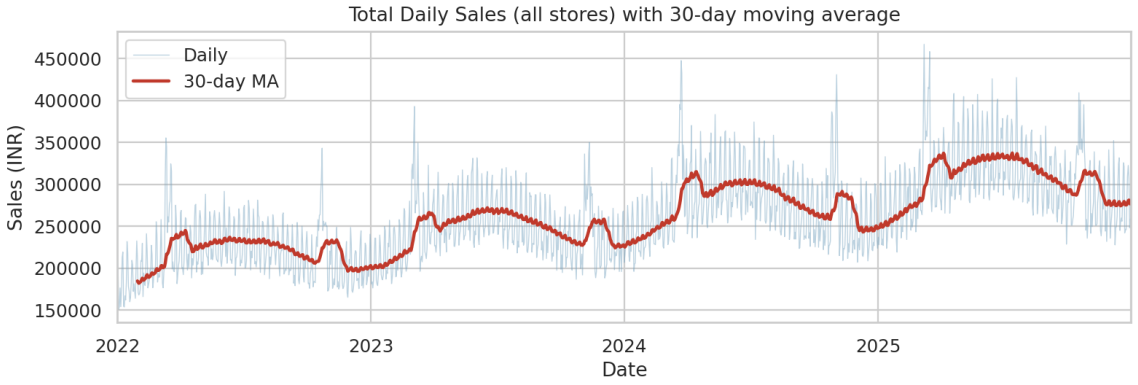


Figure 1: Total daily sales show a steady upward trend, yearly seasonality and festival spikes.



Figure 2: Target distribution, sales by store and average sales by weekday.

Observations. (i) Sales are right-skewed with a few extreme values. (ii) Lucknow is the largest store and Prayagraj the smallest. (iii) Friday-Sunday sales are clearly higher than on weekdays, with Saturday highest. (iv) Sales grow year on year; average sales are highest from March to July and lowest in January and December, with local bumps around festivals in October-November.

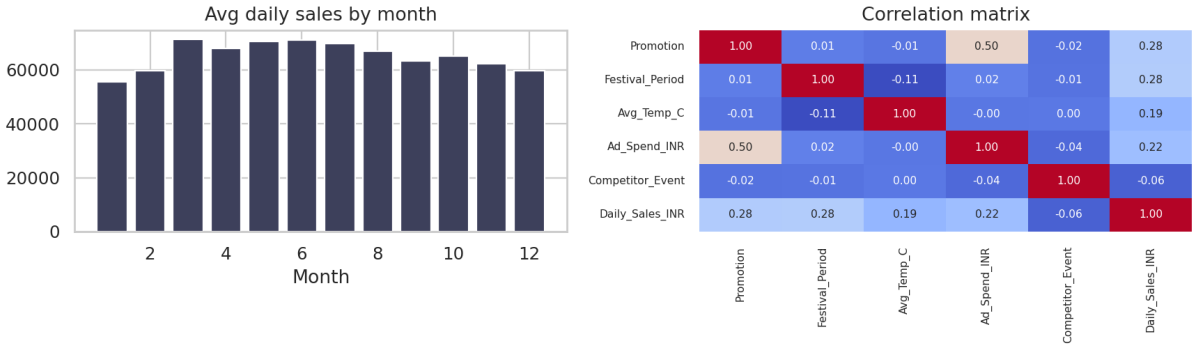


Figure 3: Monthly pattern and correlation matrix of numeric features with sales.

Correlation and effects. Promotion (r = 0.28), Festival_Period (r = 0.28) and Ad_Spend (r = 0.22) correlate positively with sales, while Competitor_Event is slightly negative (r = -0.06). On average, promotion days sell **21% more** and festival-period days **28% more** than normal days.

9. Data Preprocessing

- **Duplicates:** 15 exact duplicate rows dropped (5,859 to 5,844 rows).
- **Missing values:** filled with the store-wise median (robust to outliers).
- **Outliers:** 14 sales values above Q3 + 3 x IQR (per store) were replaced with the store median, as they were data-entry-like spikes (about 2.8x normal).
- **Encoding:** Store one-hot encoded (Region dropped as it duplicates store information).
- **Scaling:** not required for tree-based models; Linear Regression is unaffected in prediction quality.
- **Split:** time-based, not random. Training = 2022-2024 (4,160 rows), Test = 2025 (1,460 rows), to avoid leaking the future into training.

```
df = df.drop_duplicates()
for c in ["Avg_Temp_C", "Ad_Spend_INR"]:
    df[c] = df.groupby("Store")[c].transform(lambda x: x.fillna(x.median()))

# outliers: Q3 + 3*IQR per store -> replace with store median
for s, g in df.groupby("Store"):
    q1, q3 = g.Daily_Sales_INR.quantile([.25, .75]); hi = q3 + 3*(q3 - q1)
    m = (df.Store == s) & (df.Daily_Sales_INR > hi)
    df.loc[m, "Daily_Sales_INR"] = g.Daily_Sales_INR.median()
```

Summary of data-quality actions

Issue	Found	Action	Result
Duplicate rows	15	Dropped	0 duplicates
Missing Avg_Temp_C	71	Store-wise median	0 missing
Missing Ad_Spend_INR	40	Store-wise median	0 missing
Sales outliers	14	Replaced with store median	Max sales back to normal range

Feature engineering

Lag features must be known at forecast time. For a 28-day horizon the most recent usable sales value is 28 days old, so only lags of 28 days or more were used. Calendar features (month, day of week, weekend flag, week, day of year) and cyclical sine/cosine terms of the day of year capture seasonality. The target was modelled as a **ratio to the 28-day rolling mean** (de-trending), which lets tree models handle the upward trend, and predictions were multiplied back to rupees.

```
g = df.groupby("Store").Daily_Sales_INR
df["lag_28"], df["lag_35"], df["lag_56"] = g.shift(28), g.shift(35), g.shift(56)
df["roll_mean_28"] = g.transform(lambda x: x.shift(28).rolling(28).mean())
df["Sin_Y"] = np.sin(2*np.pi*df.DayOfYear/365); df["Cos_Y"] = np.cos(2*np.pi*df.DayOfYear/365)
train, test = mdl[mdl.Date < "2025-01-01"], mdl[mdl.Date >= "2025-01-01"]
model.fit(Xtr, ytr / train.roll_mean_28)          # de-trended target
pred = model.predict(Xte) * test.roll_mean_28     # back to INR
```

10. Predictive Model Development

Four regression algorithms were chosen to cover simple to advanced approaches:

Model	Why suitable
Linear Regression (baseline)	Simple, interpretable benchmark; shows how much a linear combination of features can explain.
Decision Tree (max_depth=8)	Captures non-linear rules such as 'weekend + festival = high sales'; easy to explain but can overfit.
Random Forest (250 trees, max_depth=14)	Averages many trees; reduces variance, handles interactions, robust to noise.
Gradient Boosting (400 trees, lr=0.05)	Builds trees sequentially to correct errors; often highest accuracy on tabular data.

```
models = {
 "Linear Regression": LinearRegression(),
 "Decision Tree": DecisionTreeRegressor(max_depth=8,
                      min_samples_leaf=10, random_state=42),
 "Random Forest": RandomForestRegressor(n_estimators=250, max_depth=14,
                      min_samples_leaf=3, n_jobs=-1, random_state=42),
 "Gradient Boosting": GradientBoostingRegressor(n_estimators=400,
                      learning_rate=0.05, max_depth=4, subsample=.8,
                      random_state=42)}
for name, m in models.items():
    m.fit(Xtr, ytr / btr); preds[name] = m.predict(Xte) * bte
```

Final feature set (20 inputs)

Promotion, Festival_Period, Avg_Temp_C, Ad_Spend_INR, Competitor_Event; lag_28, lag_35, lag_56, roll_mean_28; Month, DayOfWeek, DayOfYear, Week, Is_Weekend, Sin_Y, Cos_Y; and four store indicators (Store_Chennai, Store_Lucknow, Store_Prayagraj, Store_Pune).

11. Model Evaluation

As this is a regression task, models were evaluated on the unseen 2025 data using MAE, MSE, RMSE, R^2 and MAPE (mean absolute percentage error, added because it is easy for business users to interpret).

Model	MAE (Rs.)	MSE	RMSE (Rs.)	R^2	MAPE %
Linear Regression	4,953	45,730,040	6,762	0.787	6.43
Decision Tree	4,798	44,866,514	6,698	0.791	6.36
Random Forest	4,663	45,391,765	6,737	0.788	6.13
Gradient Boosting	4,812	48,742,245	6,982	0.773	6.21

Interpretation. The Random Forest is wrong by about Rs. 4,663 per store-day on average (about 6.1% of sales), and explains roughly 79% of the variation in daily sales on unseen data. The remaining ~21% is day-to-day random variation that no calendar or lag feature can explain.

12. Model Comparison

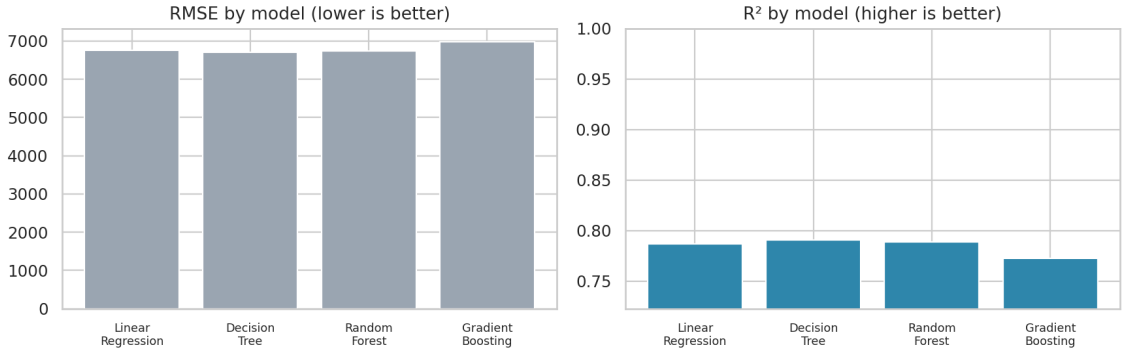


Figure 4: RMSE and R^2 of the four models on the 2025 test set.

The four models are very close: RMSE ranges only from Rs. 6,698 to Rs. 6,982. This shows that the data are dominated by a few clear patterns (weekday, festival, promotion, seasonality) that even Linear Regression can capture. **Random Forest was selected** because it has the lowest MAE and MAPE, is not dominated on RMSE (within 0.6% of the best), and is more stable than a single Decision Tree. Gradient Boosting did not improve on it, probably because the data contain substantial random noise that boosting starts to fit.

13. Final Predictions and Results

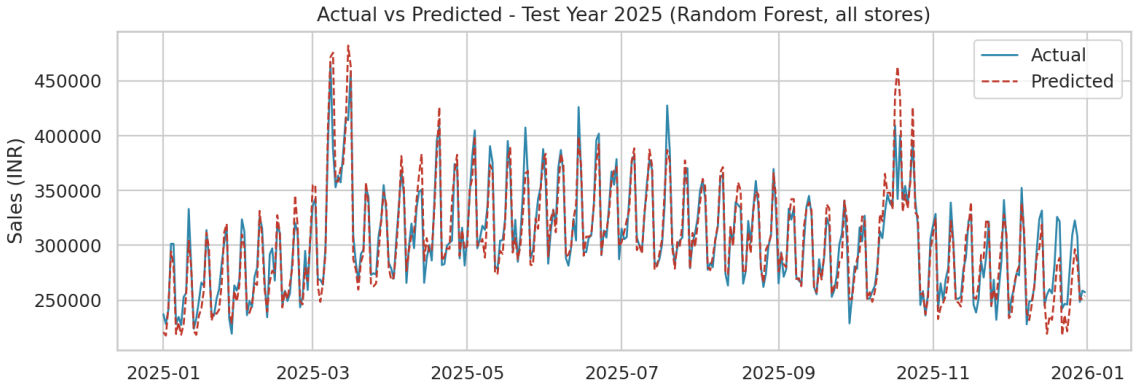


Figure 5: Actual and predicted total daily sales across the 2025 test period.

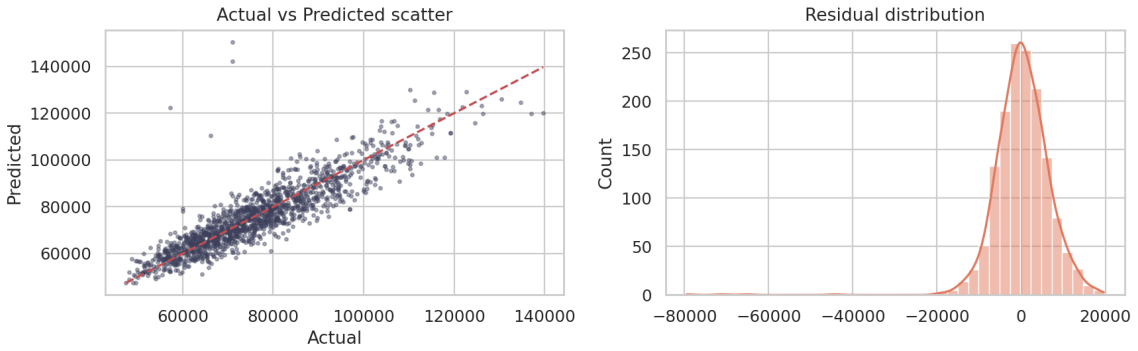


Figure 6: Predicted vs actual (points near the red line are accurate) and residual distribution (centred near zero).

Date	Store	Actual (Rs.)	Predicted (Rs.)	Error %
2025-02-24	Prayagraj	52,226	56,380	+8.0
2025-05-21	Chennai	58,840	67,390	+14.5
2025-07-23	Lucknow	104,718	97,437	-7.0
2025-08-09	Prayagraj	79,337	80,008	+0.8

Date	Store	Actual (Rs.)	Predicted (Rs.)	Error %
2025-08-19	Pune	73,000	70,013	-4.1
2025-09-06	Chennai	73,897	79,720	+7.9
2025-09-23	Pune	67,076	65,374	-2.5
2025-12-20	Prayagraj	76,577	64,439	-15.9

The predictions follow the weekly rhythm and the festival peaks, and the residuals are roughly symmetric around zero, so the model is not systematically over- or under-forecasting. Store-wise MAPE: Chennai 6.1%, Lucknow 6.6%, Prayagraj 6.1%, Pune 5.7%.

Forecast for January 2026 (28 days)

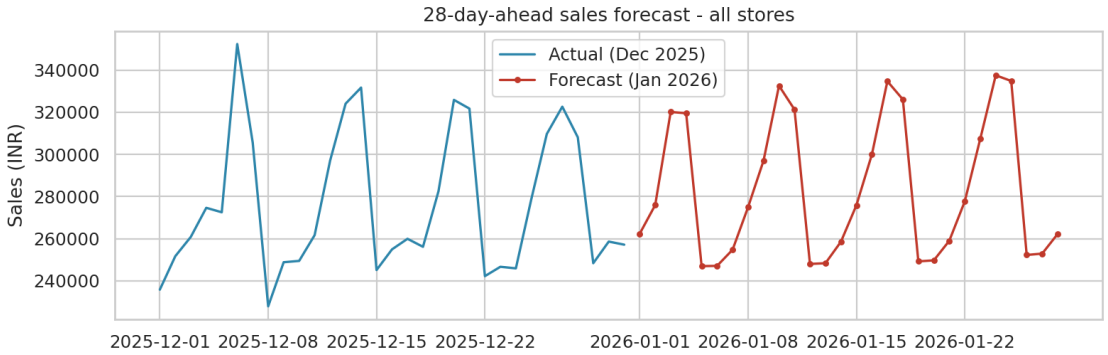


Figure 7: Forecast for 1-28 January 2026 (no promotions or festivals assumed).

Store	Forecast 1-28 Jan 2026 (Rs.)
Chennai	1,855,489
Lucknow	2,202,225
Prayagraj	1,795,037
Pune	2,074,756
Total	**7,927,506**

The model forecasts Rs. 7.93 million for the 28 days, about 8.6% above the same period in 2025 (Rs. 7.30 million), consistent with the observed growth of 10.1% in annual sales from 2024 to 2025. The full day-by-day forecast is in *forecast_jan2026.csv*.

14. Business Insights and Applications

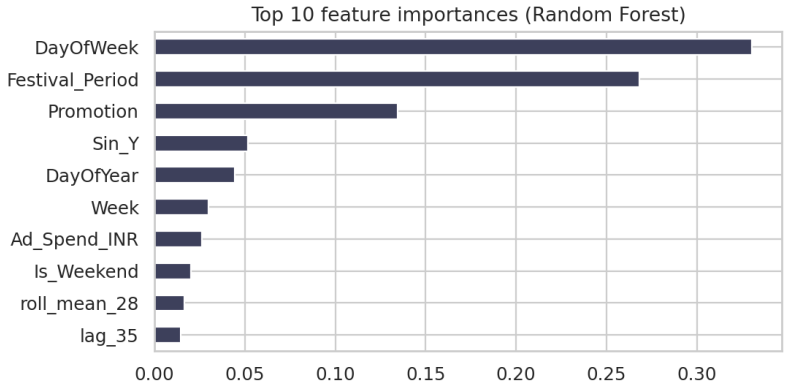


Figure 8: Top 10 drivers of sales (Random Forest feature importance).

- **Day of week is the strongest driver** (33% importance), followed by **festival periods** (27%) and **promotions** (13%). Together they explain most of the predictable variation.
- **Promotions lift sales by about 21%** and festival windows by about 28%, so promotions placed on weekday troughs can even out demand.
- Advertising spend has a modest effect; budget should be linked to measured returns.

How the forecast can be applied

- **Inventory:** order quantities four weeks ahead from store-wise forecasts, with extra safety stock for festival weeks.
- **Staffing:** build weekly rosters around the predicted Friday-Sunday peaks.
- **Marketing:** schedule promotions on predicted low-demand days; avoid overlapping them with festivals when stores are already busy.
- **Finance:** use the forecast for monthly revenue targets and cash-flow planning; set store targets with a +/-6% tolerance band.

15. Limitations and Future Scope

Limitations

- The dataset is simulated; real data would contain additional irregularities (stock-outs, returns, new product launches).
- Only four stores and four years; festival dates shift every year and must be supplied in advance.
- Future promotions, weather and competitor events are unknown at forecast time, so the January forecast assumes none; actual sales will differ if promotions are run.
- Forecast accuracy (MAPE about 6%) is limited by random daily variation; accuracy for individual days can be worse (errors above 10% occur).

Future scope

- Add real data: product category, price, footfall, inventory, local events and holiday calendars.
- Try time-series models (SARIMA, Prophet) and deep learning (LSTM), and tune hyper-parameters with time-series cross-validation.
- Add prediction intervals to express forecast uncertainty.
- Deploy as a Streamlit or Power BI dashboard refreshed monthly.

16. Conclusion

This project built a 28-day-ahead sales forecasting model for Zenith Retail. After cleaning the data and engineering calendar and lag features, four regression models were compared on a hold-out year. The Random Forest performed best on MAE (Rs. 4,663), with a MAPE of 6.1% and R^2 of 0.79. Weekday, festivals and promotions were the key demand drivers. The resulting forecast of Rs. 7.93 million for January 2026 gives the company a data-backed basis for inventory, staffing and promotional decisions, and the pipeline can be reused on real data with little change.

17. References

- Scikit-learn documentation: https://scikit-learn.org/stable/
- Pandas documentation: https://pandas.pydata.org/docs/
- Matplotlib and Seaborn documentation: https://matplotlib.org, https://seaborn.pydata.org
- Hyndman, R.J. and Athanasopoulos, G., *Forecasting: Principles and Practice* (3rd ed.), OTexts: https://otexts.com/fpp3/
- Rossmann Store Sales (Kaggle), a real-world retail dataset suited to the same method: https://www.kaggle.com/c/rossmann-store-sales
- Dataset and code: zenith_retail_sales.csv, sales_forecasting_pipeline.py (included)